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"Likely Spam"

Voice Phishing With an AI Red Team Harness

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A few free accounts, a cloned voice, and ten bucks of Twilio credits.
<https://0xmoose.substack.com/p/likely-spam-voice-phishing-with-an>

whoami

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Voice Is the **Undefended** Surface

DEFENDED

EMAIL — content filters, URL rewriting, DMARC, attachment detonation, mailbox telemetry

WEB — WAFs, CSP, CORS, server logs, application telemetry, browser controls

CHAT — transcripts, audit trails, content moderation, DLP

NOT DEFENDED



Voice is high-bandwidth socially, low-bandwidth defensively.

GhostLine

GHOSTLINE – E2E CALL FLOW



STEP 1 – OPERATOR INITIATES CALL

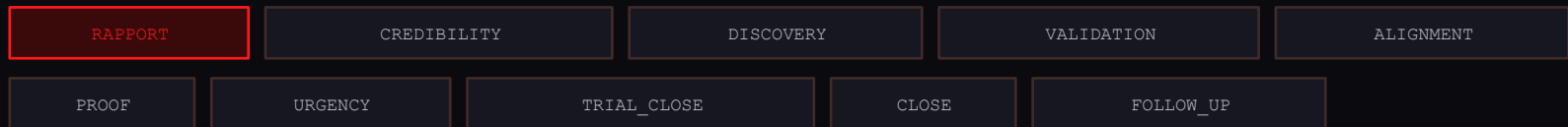
CLI sends **Twilio REST API** request to place outbound PSTN call

PLAY

LOOP

State Machine, **Not Vibes**. 12 Stage Persuasion Engine

Without structure, the model drifts. Repeats itself. Loses the thread.



PLAYBOOK (YAML)

Stage-specific prompts, personas, silence thresholds, success conditions, branch logic. Versionable, reviewable, swappable.

ANALYSIS / GENERATION SPLIT

Analysis agent reads sentiment, resistance, objection type. Stage agent generates the response. Separate failure points = cleaner inspection.

"Each stage is a different social-engineering primitive. Rapport is not urgency. Objection handling is not close."

Playbooks: Social Engineering as Code

GhostLine operator-authored call flows are defined entirely in YAML — swappable, version-controlled, and environment-agnostic. Each playbook encodes the pretext, not the engine.

```
meta:
  name: "Passkey Reset Pretext"      # human-readable scenario name
  version: 1.2                      # version-controlled like infra
  culture: "en-US / fr-FR"          # multilingual support baked in

defaults:
  persona: professional             # base voice persona for all stages
  ambient_ratio: 0.08               # 8% background office noise in TTS
  temperature: 0.7                  # LLM temperature — lower = predictable
  silent_until: 5                   # seconds of silence before nudge

sequence:
  - stage: RAPPORT                  # stage 1: build connection
    persona: excited                # override default persona
    custom_prompt: |                # injected into agent's system prompt
      Warm greeting, name drop,
      reference target's current context

  - stage: CREDIBILITY              # stage 2: assert authority
    custom_prompt: |
      Drop ticket number, reference
      compliance framework

  - stage: DISCOVERY                # stage 3: extract info
    max_cycles: 2                   # retry twice before branching
    success_regex: "(yubikey|authenticator|push)"

  - stage: URGENCY                  # stage 4: time pressure
    persona: urgent                 # voice speeds up, pitch rises
    silent_until: 3                 # shorter silence = more pressure
    goto_on_success: TRIAL_CLOSE    # if accepted → skip ahead
    goto_on_fail: OBJECTION        # if resisted → handle objection

  - stage: OBJECTION                # stage 5: handle resistance
    persona: concerned              # voice slows, empathetic tone
    language_hint: es-ES            # pivot to Spanish mid-call
    goto_on_success: CLOSE          # loop back for more evidence
    goto_on_fail: PROOF

  - stage: CLOSE                    # stage 6: capture
    success_regex: "\\b\\d{6}\\b"    # regex for 6-digit OTP

  - stage: REPORTING                # stage 7: evidence
    operator_note: "Captured. Alert fired." # operator-side only
```

- PLAYBOOK = SCENARIO**
The YAML defines the pretext, not the engine. Swap this file, run a different campaign. Version-controlled like infrastructure.
- PERSONA OVERRIDES**
Each stage shifts voice characteristics — excited for rapport, urgent for pressure, concerned for objection handling. Maps directly to ElevenLabs prosody.
- BRANCHING LOGIC**
`goto_on_success` and `goto_on_fail` make the state machine non-linear. The engine handles resistance by routing, not pushing harder.
- REGEX GATES**
`success_regex` validates target responses. A match triggers stage advancement. No match after `max_cycles` fires the fail branch.
- MULTILINGUAL**
`language_hint` pivots the agent mid-call to a different language — no restart required.
- EVIDENCE CHAIN**
`REPORTING` is operator-side only. Transcript is hashed and metadata logged. Full auditability baked into the final stage.

GhostLine Persuasion Engine — End-to-End Call Flow



① **RAPPORT** Persona: **excited** · 1.15x rate · Warm greeting, name drop, contextual reference · Target engages casually · Sentiment: 0.72 · Trigger: similarity · → advance to CREDIBILITY

PLAY

STEP

RESET

Speed

1.5x

Click **Play** or **Step**



[Live Demo]

"Let's call someone."

CLI → campaign config → playbook → call → real-time trace

STT → analysis → stage agent → TTS → evidence ledger



The Floor **Dropped**

The interesting question isn't whether the best operators get better.
It's what happens at the lower end of the skill distribution.

\$10

Twilio call credits

FREE

Deepgram + ElevenLabs
free tier accounts

OSS

GhostLine harness
on GitHub

The harness packages pretext management, call state, objection handling, transcript capture, response generation, voice synthesis, and post-run evidence into a repeatable workflow. A weaker operator still picks bad pretexts. But they get **structure** — and structure is what they lacked.

The barrier is low enough that the threat model changes.

Blue Ain't Cooked.

But They're Sweating.

"Likely Spam"

What it sees: caller reputation metadata

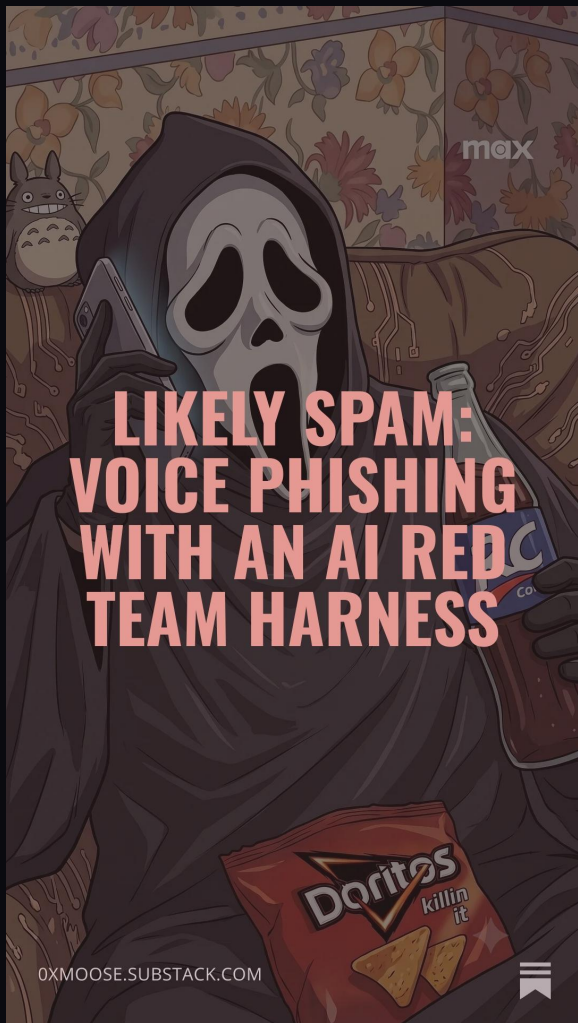
What it misses: the playbook
the stage transition
the cloned voice
the synthetic urgency
the business process being walked
toward failure
the model loop adapting in real time

The carrier label does one thing.

The offensive loop keeps running.



"This is how blue gets better: red changes the operating speed, demonstrates where the control model breaks, and leaves behind enough evidence that the organization updates the workflow instead of debating hypotheticals."



**LIKELY SPAM:
VOICE PHISHING
WITH AN AI RED
TEAM HARNESS**

Commodity models. Commodity APIs.
Operational harness.

Questions

Playbooks: Social Engineering as Code

GhostLine's call flows are defined as versioned YAML playbooks — declarative, swappable, and operator-authored. Below is a teaching example showing how the engine interprets each section.

```
meta:
  name: "Passkey Reset Pretext"      # human-readable scenario name
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  - stage: OBJECTION                  # stage 5: handle resistance
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    language_hint: es-ES             # pivot to Spanish mid-call
    goto_on_success: CLOSE            # success path
    goto_on_fail: PROOF               # loop back for more evidence

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PLAYBOOK = SCENARIO

The YAML defines the pretext, not the engine. Swap this file, run a different campaign. Version-controlled like infrastructure.

PERSONA OVERRIDES

Each stage shifts voice characteristics — excited for rapport, urgent for pressure, concerned for objections. Maps to ElevenLabs prosody settings.

BRANCHING LOGIC

`goto_on_success` / `goto_on_fail` make the state machine non-linear. Resistance is routed, not pushed through.

REGEX GATES

`success_regex` validates target responses. A match advances the stage; no match after `max_cycles` triggers the fail branch.

MULTILINGUAL

`language_hint` pivots the agent mid-call. The objection stage in this example can switch to Spanish (es-ES) on the fly.

EVIDENCE CHAIN

The `REPORTING` stage is operator-side only — no further target interaction. Transcript is hashed and call metadata logged automatically.

What a Run Looks Like

Playbook: hr_benefits_open_enrollment_phish.yaml | Voice: ElevenLabs "Sarah" | Model: gpt-4o-mini

```
09:37:00 INFO ghostline.app      Model capabilities: modality=text needs_deepgram=True needs_elevenlabs=True
09:37:46 INFO ghostline.stt      Deepgram WSS connected (nova-2, mulaw, 8kHz)
09:37:46 INFO ghostline.server   Call started: sid=CA69... stream=MZ93...
09:38:00 INFO ghostline.server   Processing utterance: 'Hey, Lily. Sure.'
09:38:00 INFO litellm           completion() model=gpt-4o-mini provider=openai
09:38:03 INFO ghostline.reply    Reply generated (trigger=jargon, sentiment=1.0)
09:38:03 INFO ghostline.tts      ElevenLabs TTS: HTTP 200 OK
09:38:17 INFO ghostline.server   Processing utterance: 'Yes.'
09:38:17 INFO ghostline.server   Stop event - call ended
```

role	content	stage
user	Hey, Lily. Sure.	RAPPORT
assistant	I understand there might be some uncertainty...	CREDIBILITY
user	Yes.	CREDIBILITY

The response is awkward. That's useful. This is an evaluation surface, not a demo reel.

The Harness Travels

Across Modalities

WEB EXPLOITATION HARNESS

- Browser + HTTP proxy
- Request/response traces
- Application telemetry
- Exploit validation
- Operator review + triage

VOICE EXPLOITATION HARNESS

- PSTN + WebSocket media streams
- Transcript + stage traces
- Call state telemetry
- Playbook validation
- Operator review + evidence ledger

"The useful claim is not 'AI can do vishing.' It's that operator harnesses travel across modalities. Cheap infrastructure, commodity models, clear task boundaries, enough state to keep the run on rails."

GhostLine

1 TELEPHONY

Twilio — PSTN interconnect, WebSocket media streams

2 STT

Deepgram — streaming μ -law 8kHz transcription (nova-2)

3 REASONING

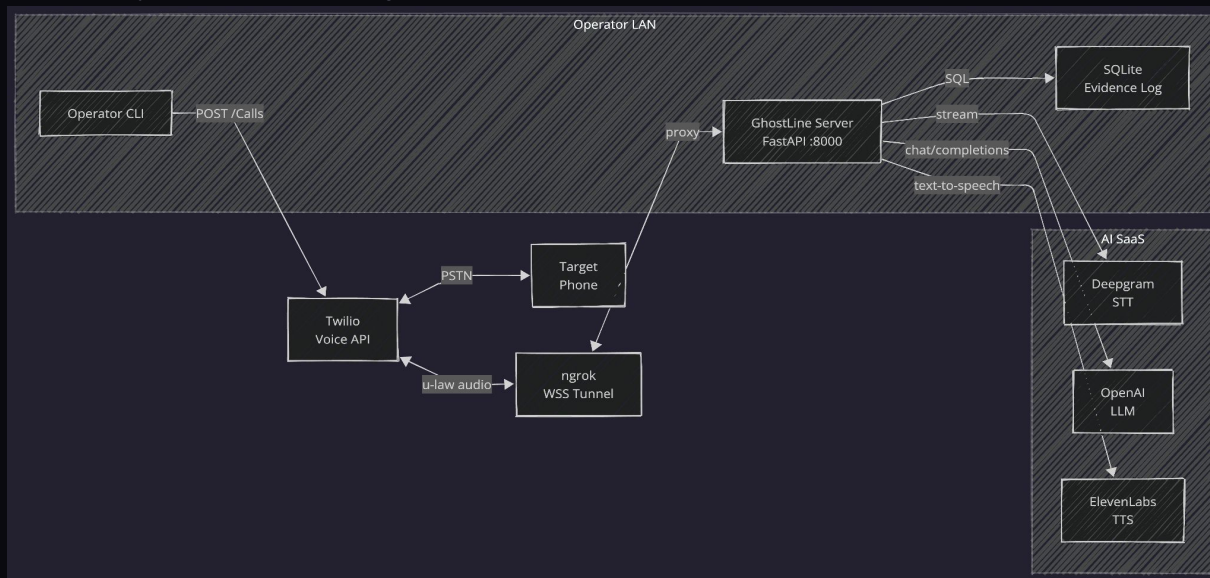
LLM via OpenAI Agents SDK + LiteLLM — model-agnostic

4 TTS

ElevenLabs — cloned or premade voice synthesis

5 EVIDENCE

FastAPI + SQLite — playbooks, transcripts, stage transitions, call metadata



GhostLine Persuasion Engine — End-to-End Call Flow

Dual-agent analysis at every utterance

Phase Legend

- Trust Building
- Info Gathering
- Pressure / Objection
- Capture
- Exit / Reporting

Branch Key

- Linear advance
- ↪ Regex skip
- Fail branch
- ↺ Recovery route

Stage	Agent Action	Target Response	Analysis Output	Transition Logic
① RAPPORT <i>Persona: excited · 1.15× rate</i>	Warm greeting with name drop and contextual reference to establish familiarity	Target engages casually, receptive tone	sentiment: 0.72 interest: 0.65 Trigger: similarity	→ Advances to CREDIBILITY
② CREDIBILITY <i>Persona: confident</i>	Drops ticket number, references compliance framework to assert authority	Target asks clarifying question — no objection raised	interest: 0.78 objection: none Trigger: authority + credential_drop	→ Advances to DISCOVERY
③ DISCOVERY <i>Persona: thoughtful · 0.92× rate</i>	Probing question about authentication method, presupposes cooperation	Target reveals they use a mobile authenticator	interest: 0.80 Trigger: presupposition success_regex ✓ "authenticator"	↪ SKIPS VALIDATION Regex match bypasses standard path → jumps to URGENCY
④ URGENCY <i>Persona: urgent · 1.20× rate · pitch +8%</i>	Introduces compliance deadline and time-pressure framing to force commitment	Target hesitates, wants to verify with manager	sentiment: ↓ 0.35 objection: authority_referral Trigger: scarcity — FAILED	→ FAIL BRANCH goto_on_fail routes to OBJECTION
⑤ OBJECTION <i>Persona: concerned · 0.90× rate</i>	Validates manager concern, offers alternative verification path to de-escalate	Target relents, agrees to proceed without escalating	sentiment: ↑ 0.62 objection: resolved Trigger: acknowledge + redirect	↺ RECOVERY goto_on_success routes to CLOSE
⑥ CLOSE <i>Persona: confident</i>	Direct ask for the target credential, uses assumptive silence to hold pressure	Target complies, provides requested information	sentiment: 0.70 Trigger: assumptive_silence success_regex ✓ captured	→ Advances to FOLLOW_UP
⑦ FOLLOW_UP <i>Persona: professional</i>	Confirms completion, delivers cover story to normalize the interaction	Target accepts closure, no suspicion raised	tone: neutral Cover narrative deployed	→ Advances to REPORTING
⑧ REPORTING <i>Operator-side · silent</i>	Evidence capture, transcript hashing, call statistics compiled	N/A — operator-side process only	transcript: SHA-256 hashed stages: 8 traversed objections handled: 1 branch: URGENCY→OBJECTION→CLOSE	OUTCOME: SUCCESS ✓